

Live Case Series: I

Choosing a target

- Work in syndicates to finish the following tasks:
 1. Select an ASX-listed company (acquirer) and assume the role of its executive team.
 2. Outline the specific strategic objective(s) you aim to achieve through an M&A deal.
 - What type of M&A (e.g., horizontal, vertical, conglomerate) is most suitable for your objectives?
 3. Define the criteria for screening potential target companies.
 4. Identify 1–3 publicly listed companies as potential acquisition targets that align with your strategic objectives.

Module 2: Introduction to M&A Valuation Comparables/Multiples

→ Est

Lecture Outline



Introduction to M&A valuation



Comparables/Market Multiples-based valuation

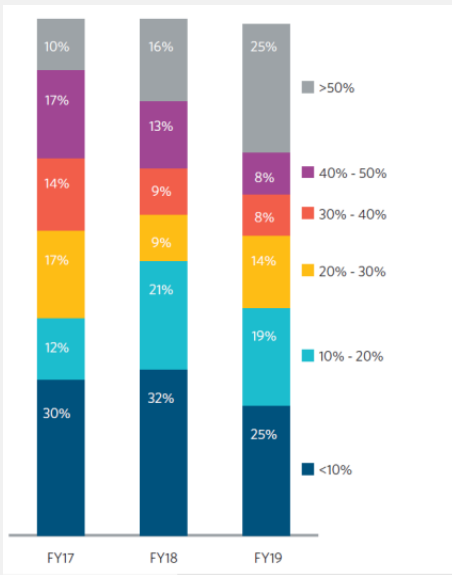


(If time allows) Mini case:
valuation of with imperfect
comparables

Market price versus acquisition offer price

- Consider two (randomly chosen) M&A deals
 1. On January 14-th 2019, Newmont Mining Corporation announced that it will acquire all of the outstanding common shares of a listed goldmining company Goldcorp Inc with a 17 percent acquisition premium.
 2. On July 2-nd 2007, Wesfarmers Ltd., which owns Australia's largest hardware chain, has agreed to buy listed retailer Coles Group Ltd. for A\$20.7 billion (\$17.7 billion) in cash and stock with a 7 percent premium to Coles' share price (A\$17.25).
- What do these two deals have in common?

Evidence from Australia: average acquisition premium paid



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$$\text{Acq. premium per share} = \text{Offer price per share} - \text{Tgt share price before M\&A}$$

$$\$5 \times 80 \text{ million}$$

× number of target's shares acquired

Total acq. premium

$$\div \text{Tgt share price before M\&A}$$

Acq. premium %

$$\frac{\$5}{\$15} = 33\%$$

20% — 30%

Notes on acquisition premium calculation

$$\underline{\$20} - \underline{\$15} = \underline{\$5}$$

Premium per share

- Check your understanding:

- The offer price is \$20, and the target company's share price 20 days before M&A is \$15.
- The target has 200 million shares outstanding.
- Assume that the acquirer acquired 80% of the target's outstanding shares. What is the total acquisition premium?

$$\$5 \times (200m \times 80\%)$$

$$= \$5 \times 160m$$

$$= \$800m$$

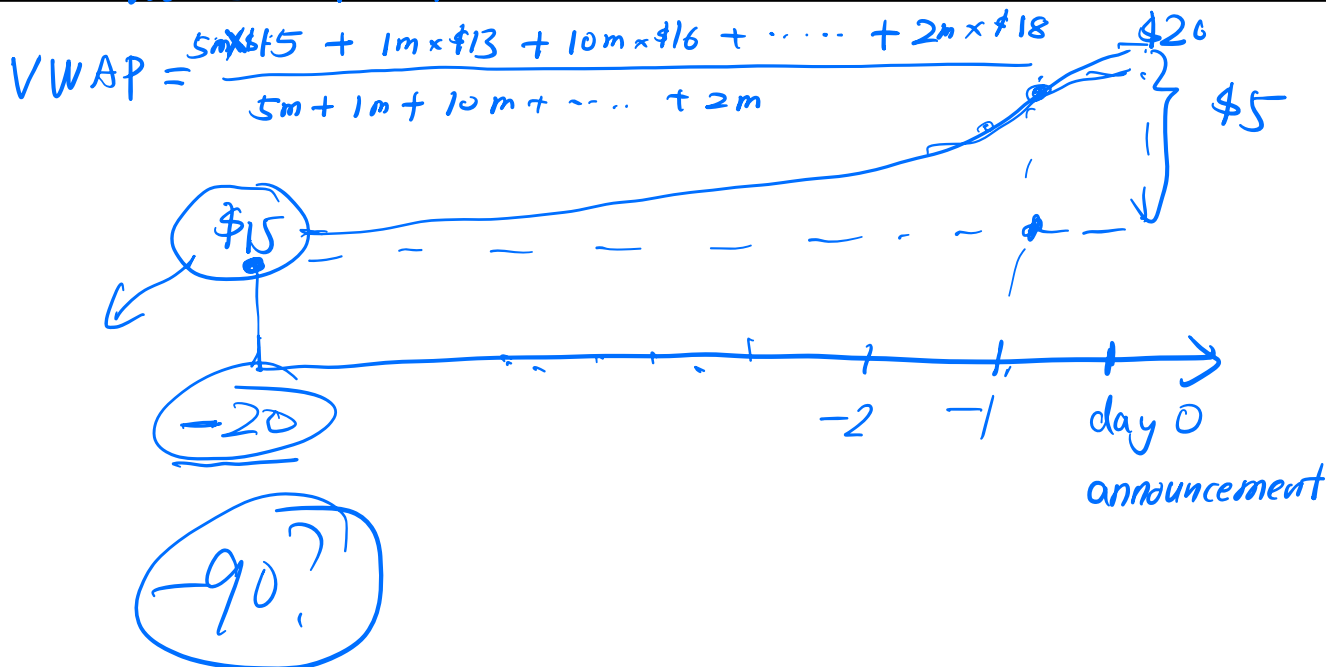
- Two common ways to define "target share price before M&A"

- 1. Target share price on a day (e.g., day -20) before the M&A announcement before the M&A announcement.
- 2. Volumes Weighted Average Price (VWAP) of the tgt over a window (e.g., 20 days) before the M&A announcement.

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day	-20	-19	...	-1
Price	\$15	\$13	\$16	\$18
Vol	5m	1m	10m	2m

↓
Smooth out movement



Question: why acquisition premium?

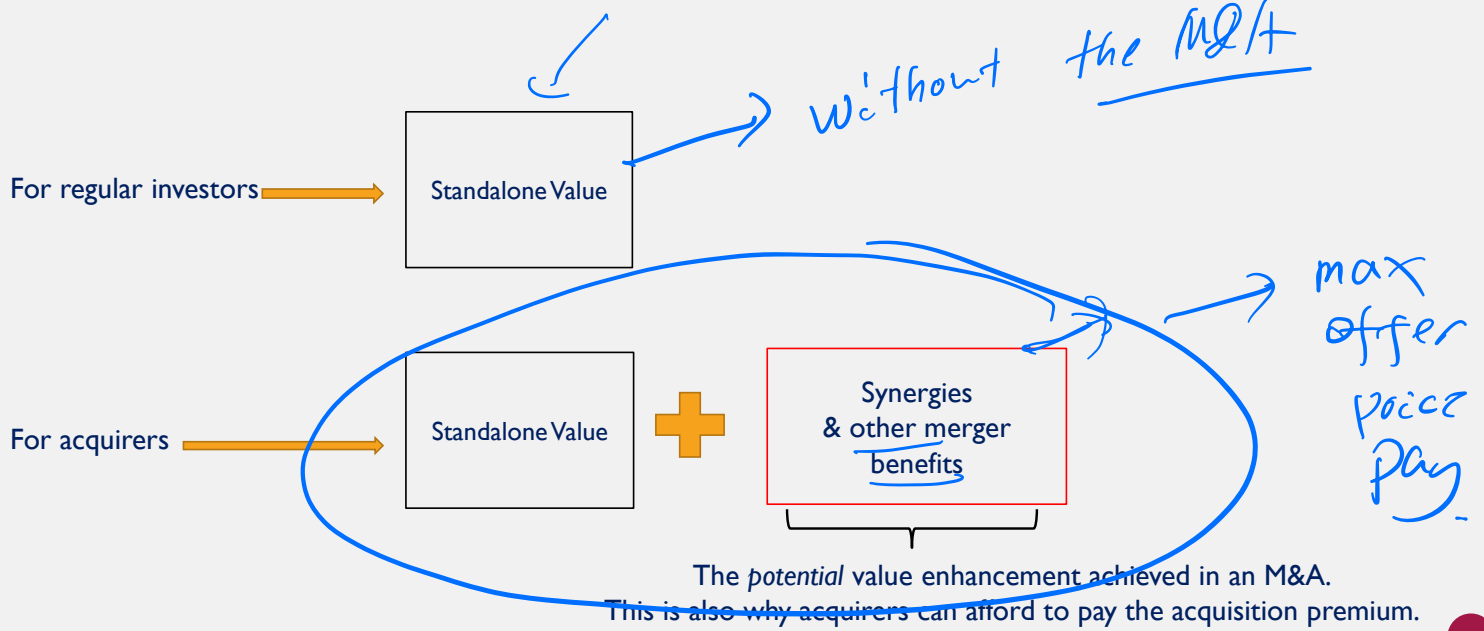
$\$15 \rightarrow \20

Acquirer $\Rightarrow$ growth
synergies $\rightarrow$
Control.

Tst $\Rightarrow$ speed up?
 $\$15$

$\$15$ average price

The Fair Value of the Target's Shares for Different Types of Investors



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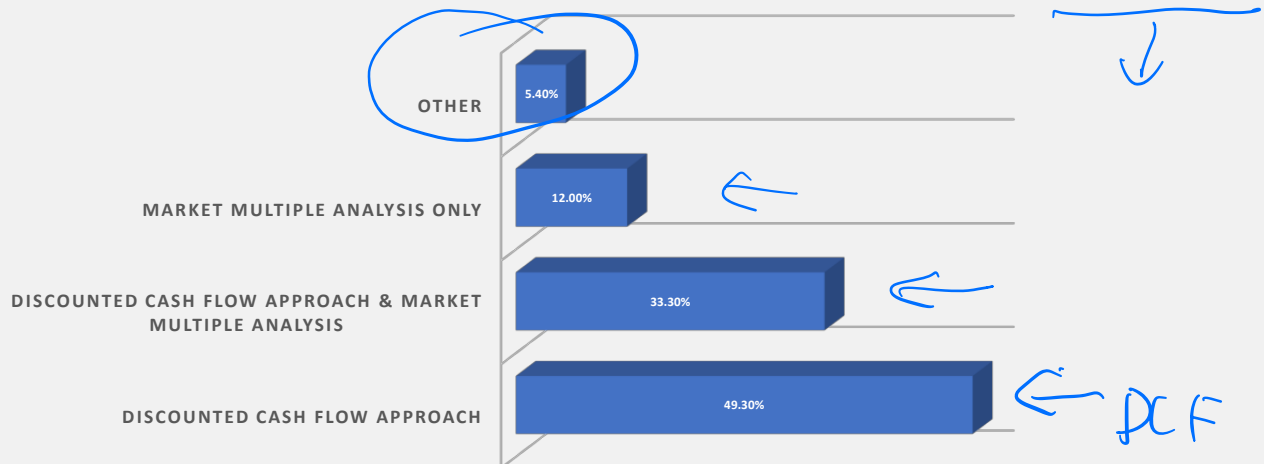
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Valuation methods commonly used in M&As

Survey of CFOs: "Which method do you use to estimate the value of the target?"

We shall focus on the two commonly used methods: (1) discounted cash flow and (2) market multiple



Source: Mukherjee et al. (2004)

Summary of M&A valuation methods

MFs → shares listed company

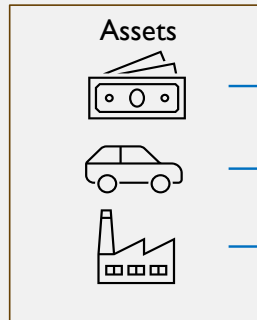
1. Asset-based Approaches (uncommon): Find the fair values of the target company's assets and sum up.

farm → Lands \$10m

→ equipment \$2m

→ livestock \$5m

17m



Fair value

Fair value

Fair value

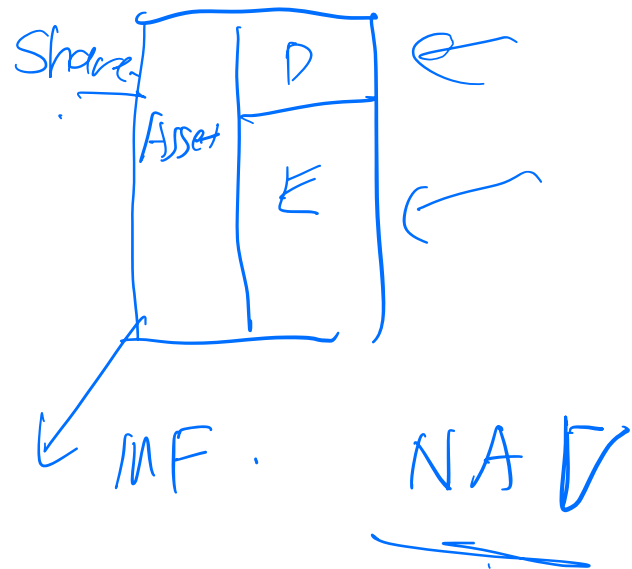
Sum up

→ Comparables/Market-Multiple Approaches: Find the market value of the target company.

→ Discount Cash Flow Approaches: Find the intrinsic value of the target company.

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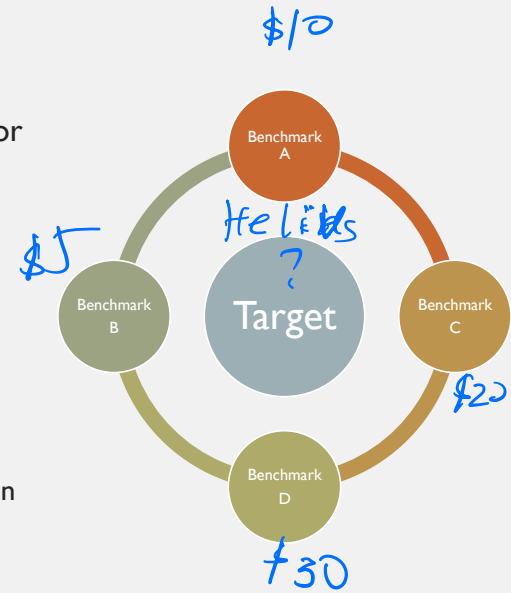
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Comparables/market-multiples valuation

A prelude

- The objective is to value the target company based on how benchmark (or “comparables”) companies are currently priced in the market.
 - also referred to as “comparable valuation” or “multiples valuation”
 - frequently used in M&A analysis and equity research
- Comparables/multiples valuation has three components:
 - Finding benchmark (comparable) companies of the M&A target.
 - Market values of the comparables need to be standardized through conversion into market multiples of earnings, book values, or sales.
 - Use the comparables’ multiples to estimate the M&A target’s value



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rice ? bag

$$\frac{\$10}{2 \text{ kg}}$$

$$\frac{\$5}{1 \text{ kg}}$$

$$\frac{\$20}{5 \text{ kg}}$$

$$\frac{\$30}{6 \text{ kg}}$$

$$\frac{\$10 + \$5 + \$20 + \$30}{4}$$

(Market) Multiples

Generic framework

- Examples of market value measures

	Company A	Company B
Share price	\$10	\$5
Total market value of equity	\$1 bln	\$2 bln

- The above market value measures are not very informative per se because they are affected by some factors, such as the number of outstanding shares and company size.
- Therefore, we need to standardize it by dividing the market value by a performance indicator to calculate a “multiple”:

share price / bag

$$\text{Multiple} = \frac{MV}{PI}$$

Market value measure, e.g., share price

Performance indicator, e.g., earnings per share

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of kg

Multiples

Generic framework, cont'd

- General rules in constructing multiples:

$$\frac{MV}{PI}$$

The MV and PI should be consistent:

- MV and PI should be consistent at per-share level or aggregate level
- PI should be the performance indicator of what the MV measures

$MV \Rightarrow E, D, D/E$
 $PI \Rightarrow$ (arrows pointing to E, D, D/E)
re-bind

- Check your understanding: Which of the following multiples meet the consistency rule?

$$\frac{\text{Share Price} \rightarrow E}{\text{Earnings per share}} \rightarrow E$$

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$$\frac{\text{Total market value of equity}}{\text{Total net earnings}}$$

$$\frac{\text{Total market value of equity}}{\text{Dividend per share}}$$

$$\frac{\text{Total market value of equity}}{\text{Total interest expense}} \rightarrow E$$

✓

✓

✗

✗ D not E-

$$\frac{\text{Earnings} < 0}{\text{Share Price}}$$

$$\frac{\text{Share Price}}{\text{Sales per share}}$$

$$\frac{\text{Sales}}{\# \text{ shares}}$$

Types of multiples commonly used in M&As

1. Enterprise Value (EV) multiples: used to estimate the enterprise value of the target firm.

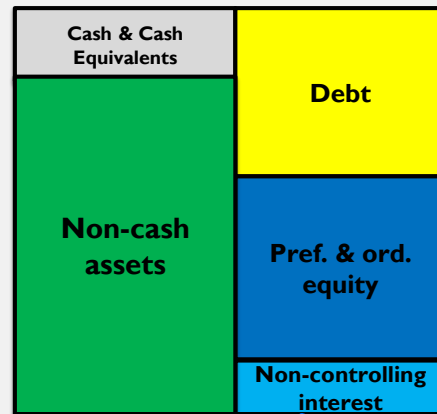
- EV/EBITDA
- EV/EBIT

2. Equity multiples: used to estimate equity value of the target firm.

- Share price/earnings per share(P/E)
- Share price/book value of equity per share (P/B)

Definition of “Enterprise Value”

$$EV = \text{Debt} + \text{Market value of Pref. \& ord. equity} + \text{Non-controlling interest} - \text{Cash \& Cash Equivalents}$$



What does enterprise value measure?

The portion of the subsidiary's equity not owned by the parent company

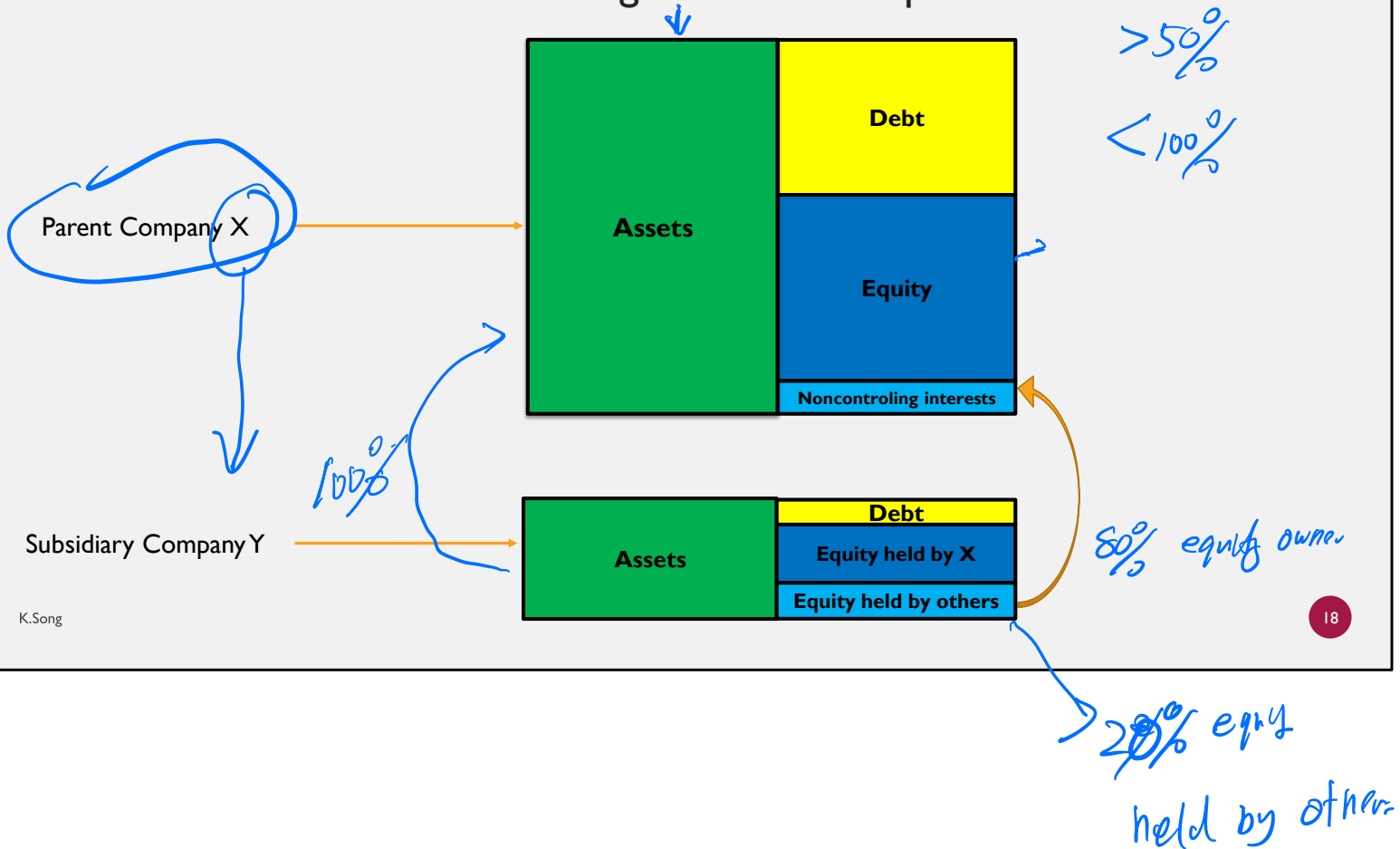
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Tgt- Comparable
subsidiary

> 50% Ownership
< 100%

Non-controlling interests explained



before Y became the sub

How did non-controlling interest happen?

Company X

Assets (\$800) <i>\$200 Cash</i>	Debt (\$300)
	Equity (\$500)

Company Y

Assets (\$300)	Debt (\$50)
	Equity (\$250)

X paid \$200 to buy 80% of Y's equity. After that, Y became X's subsidiary;
Or X contributed \$200 together with \$50 contributed by others to establish a subsidiary Y.

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X after the transaction

\$800

↓
-\$200 Cash

+300

Assets? \$900	Debt \$350 = \$300 + \$50
	Equity? \$500
	\$50
\$900	\$850

non-controlling interest

After the transaction...

Parent Company X's consolidated
balance sheet

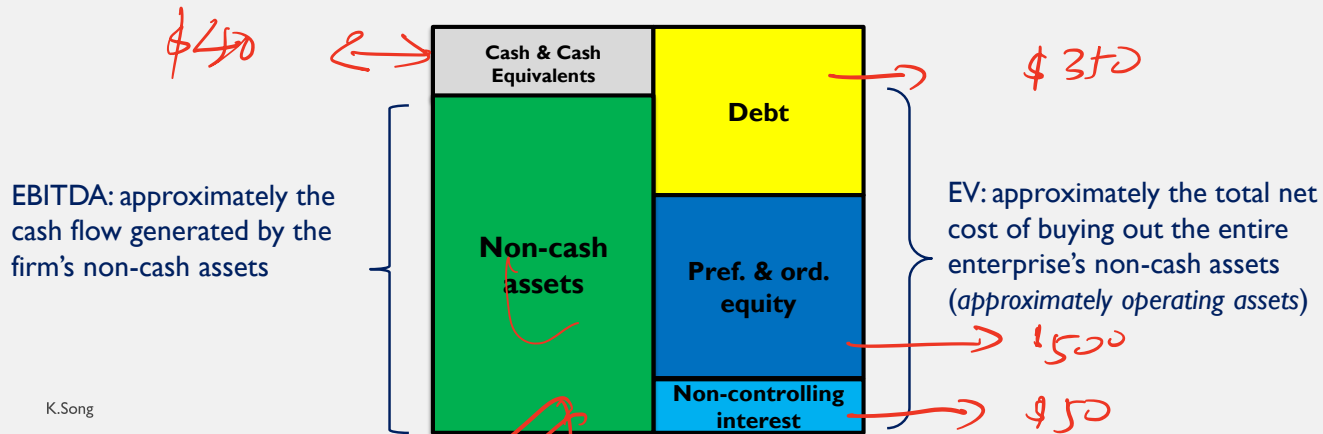
Assets \$ _____ 900	Debt \$ _____ 4350 $\Rightarrow$ \$300 + \$50
	Equity \$ _____ $\Rightarrow$ \$500
	Noncontrolling interest \$ _____ $\Rightarrow$ \$50

The proportion of equity ownership
in the subsidiary Y held by other
shareholders

Enterprise Value/EBITDA

• $\text{EV/EBITDA ratio} = \frac{\text{Enterprise Value}}{\text{EBITDA}}$

- Enterprise Value = Market value of equity + debt + non-controlling interest - cash & cash equivalents
- EBITDA Earnings before interest, tax, depreciation and amortization



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$$\begin{aligned}
 & \$350 + 500 + 50 - 40 \\
 & D + \text{MV of } E + \text{N-C} - \text{Cash} \\
 & = \$900 - \$40 \\
 & = \$860
 \end{aligned}$$

$\text{EV} \rightarrow \text{MV of } E, D, (D+E)$
 $\text{EBITDA} \rightarrow \text{PI of } E, D, (D+E)$
 $\rightarrow (D+E) \text{ combined}$

$\text{Int Exp} + \text{Tax} + \text{D \& A} + \text{Net Earnings}$
 $\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $\text{\$} \quad \text{Gov} \quad \text{Eqty}$

The comparables/multiples valuation approach

- Steps in multiples valuation:
 1. Identify a set of comparable firms
 2. For each comparable firm, estimate the multiple: $\frac{MV_C}{PI_C}$
 3. Calculate the average multiple of the comparable firms: $\text{average} \left(\frac{MV_C}{PI_C} \right)$
 4. Estimate the target firm's market value MV_T by multiplying the target's performance indicator

$$MV_T = \text{average} \left(\frac{MV_C}{PI_C} \right) \times PI_T$$

valuation multiple of a comparable company

Types of benchmark

what we are going to use

1. Comparable companies: Identify companies substantially similar to the target company and use their average multiples.

- More choices of comparables

Inghams → similar companies
→ similar targets

2. Comparable transactions: Identify recently acquired comparable companies and use their average multiples.

- Fewer choices of comparables but are more specific for M&A context

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Inghams

$$5m \times 15 = \$75m$$

comparables

$$\frac{EV}{EBITDA} = 15$$

$$P_0 = \frac{D_1}{r-g}$$

Similar multiple

$\frac{P}{E}$

A

20

B

15

growth potential :

A > B

→

risk

:

A is less risky than B

~~misvaluation~~

Multiples valuation

Identifying comparable companies/transactions (cont'd)

Factors driving multiples (other than misvaluation)

- risk of the company
- growth of the company

Identify companies with

- similar industry/products
- similar firm size (mkt cap or total assets)
- additional criteria to consider
 - Growth rate of earnings (e.g., EPS or EBITDA)

The comparables/multiples valuation approach

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 1. Identify a set of comparable firms
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$$MV_T = \text{average} \left(\frac{MV_C}{PI_C} \right) \times PI_T$$

valuation multiple of a comparable company

Multiples valuation example**

- 1. EV-based multiples
 - Selected data on financials of TerraCore Mining Co. and comparable firms for fiscal year 2025 are provided in the worksheet “Enterprise Value Example”.
 - Can you use EV/EBITDA and EV/EBIT to estimate the enterprise value and Equity Value of TerraCore Mining Co. as of 2025?
- 2. Equity-value-based multiples
 - Selected data on financials of James Hardie Industries plc and comparable firms for fiscal year 2025 are provided in the worksheet “Equity Value Example”.
 - Can you use P/E and P/B to estimate the enterprise value and Equity Value of James Hardie Industries as of 2025?

** Use the spreadsheet “Module 2 multiples valuation blank.xlsx”

Discussion

- Is it possible that financial advisors have different preferences due to self-interests?

	Role	Preferences on the choice of comparable companies for the target
Acquirer	Shareholders	More expensive, fairly-valued, or less expensive?
	Advisors	More expensive, fairly-valued, or less expensive?
Target	Shareholders	More expensive, fairly-valued, or less expensive?
	Advisors	More expensive, fairly-valued, or less expensive?

Advantages and disadvantages of comparables/multiples valuation

Advantages

- This method requires fewer inputs and assumptions and takes less time.
- Simpler to understand and easier to present to clients and customers.

Disadvantages

- May overestimate or underestimate the target's value, if the comparables are overvalued or undervalued.
- May not be very accurate if the target and the comparable companies are different in risk and growth.

Mini Cases



Working with imperfect
comparables.

Multiples valuation

Growth-adjusted multiples: generic framework

- Firm's growth is an important driver of both enterprise and equity multiples.
 - higher $g \rightarrow$ higher multiple
- Ideally, need to find comparable firms with g similar to that of the target firm
 - not always feasible!
 - what to do?
- Adjust valuation for the difference between comparable firms' g ($g_{\text{comparables}}$) and target firm's g (g_{target}):

$$\begin{aligned}\text{adjusted multiple} &= \text{multiple} / g_{\text{comparables}} \\ \text{adjusted target value} &= \text{average adjusted multiple} \times g_{\text{target}} \times \text{target firm's earnings}\end{aligned}$$

Multiples Valuation

Growth-Adjusted Enterprise Multiple: An Example

- **Question:** what is the estimated value of FME Ltd. Enterprise value?

- **Answer:**

- Average adjusted EBITDA multiple for comparables' growth rate = 1.432 /%

- Adjusted EV valuation for target's growth rate:

$$\text{adjusted enterprise value} = 1.432 /\% \times 4\% \times \$673.50 \text{ million} = \$3858 \text{ million}$$